Timofei Gorchakov

ALGORITHMIC PROBLEM DESIGN PORTFOLIO

I have a long background in mathematical problem solving. I studied in the mathematics track at Moscow School 57 and later studied Mathematics at HSE University. I participated in mathematical olympiads throughout my school years and spent about eight years tutoring olympiad mathematics, beginning during my last two years of school and continuing for six years afterward.

I later spent about five years as an analyst at Yandex, primarily working with econometrics, modeling, and time series. Alongside my professional work, I kept returning to algorithmic problem solving.

Over time, I became increasingly interested not only in solving algorithmic problems, but in **designing them**. After seeing enough problems, I found myself asking not only "How do I solve this?" but also "Why is this a good problem?" - what makes an insight surprising but discoverable, how constraints reveal the intended structure, and how an apparently complicated search space can collapse after choosing the right representation.

I had designed mathematical olympiad problems before, and I wanted to see whether I could apply the same creative process to algorithmic problems in the style of LeetCode. I created this portfolio as a focused problem-design project and as a way to demonstrate the kind of work I would like to do professionally.

The six selected problems came from a substantially larger pool. During the same two-week project, I developed roughly 15-20 additional draft ideas and excluded them when the central insight was not strong enough, the problem was too close to existing work, or the overall design did not meet my bar.

AT A GLANCE

8 years Olympiad math tutoring

5 years Analyst at Yandex

1,150+ LeetCode problems solved

1,925 LeetCode contest rating

Top 4% in LeetCode contests

WHAT I CAN CONTRIBUTE

- Problem ideation & selection
- Difficulty calibration
- Solution design
- Correctness & complexity proofs
- Edge-case analysis
- Editorial writing
- Prior-art review

INTERESTED IN

Problem Developer | Algorithmic Content Engineer | Problem Setter / Reviewer

SELECTED PROBLEMS

Problem	Difficulty	Core idea
Maximum Cylindrical Shift Score	Hard	Global range → fixed cyclic distance → independent ring optimization
Palindromic Grid Paths	Hard	Palindrome → half-walk + center → multinomial counting
Minimum XOR with Binary Tree Path Values	Hard	Query bound → only 20-bit suffixes matter → greedy binary trie
Minimum Variance Reconstruction	Medium	Adjacent sums → one scalar parameter → exact convex optimization
Two Targets on a Billiard Table	Hard	Unfolding → finite abelian group → cyclic subgroup criterion
Longest Path in a Prefix Graph	Hard	Dense prefix graph → nearest-prefix forest → DP

Full portfolio: Algorithmic Problem Collection - six independently designed problems with complete statements, examples, editorials, correctness arguments, and complexity analyses (26 pages).

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Open to ongoing part-time/hourly work and contract-based problem development